Eklavya Gupta

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Product Engineer with hands-on experience in LLM agents, hybrid search, and backend optimization. Seeking backend or AI engineering roles targeting meaningful performance and product outcomes.

SKILLS

Programming Language: Python, Node.js, TypeScript, JavaScript, Rust, SQL

Development: HTTP, REST APIs, Database Management, Web Sockets, Caching, SOLID

DevOps: AWS - (Lambda, EC2, S3, AgentCore), Podman, Docker, Microservices, CI/CD, GitHub Actions, Jenkins

Others: Git, OOPs, System Design - (LLD, HLD), Agent, Langgraph, Langchain, RAG, MCP (Model Context Protocol), LLM Domain Adaption/Fine Tuning

Frameworks & Libraries: Node.js, Express.js, FastAPI, FastMCP, Prisma, PostgreSql, MongoDb, Socket.io, Redis

WORK EXPERIENCE

Product Engineer **Nov 2025 - Present**
Tata Consultancy Services **Noida, India**
ADD Safety - LifeSciences

- Architected and deployed a **LangGraph-based AI agent** on AWS AgentCore Runtime with integrated short-term conversational memory, enabling stateful, context-aware interactions across multi-turn sessions in the LifeSciences domain.
- Designed and implemented a **hybrid retrieval pipeline** (BM25 + Semantic Search → Stage-1 Reranking → Stage-2 ACE Reranking) for medical coding queries within an MCP server, improving top-k retrieval accuracy from **~60% to 95%+**.
- Led experimentation using **ACE (Agentic Context Engineering)** — a domain adaptation framework comprising Generator, Reflector, and Curator stages — achieving **80% accuracy** in medical coding, driving measurable gains in specialized LifeSciences NLP tasks.
- **Optimized the ACE training pipeline by 3–5×**, reducing end-to-end domain adaptation time from 24 hours to **5–6 hours** for equivalent record volumes through script-level parallelism and compute optimizations.
- **Reduced full inferencing pipeline runtime by 75%** (8 hours → 2 hours) by introducing concurrency at database and network call layers and applying result caching on static/deterministic functions within the MCP server pipeline.
- Automated end-to-end CI/CD for the agent and MCP server using **Jenkins**, eliminating manual deployment overhead and standardizing release workflows.
- Built an **AWS Lambda-based memory snapshot service** to back up AgentCore conversational memory, ensuring data durability and recovery for production agent sessions

Backend Engineer & Machine Learning **Nov 2024 - Nov 2025**
Infigos Software Solutions **Remote**
US Government RFP Proposal Automation

- Developed a production-ready **FastAPI** application with **async/await** patterns, implementing RESTful APIs for **content generation** and document processing. Integrated **multiple AI services** (OpenAI, Anthropic) with **vector database** (**Pinecone**) for semantic search and knowledge retrieval.
- Architected a modular **backend system with 12+ specialized services** including content generation, compliance checking, and knowledge base management.
- Implemented **dependency injection** and service layer patterns for maintainable code structure.
- Implemented structured data validation using Pydantic schemas and built **custom tool execution** framework for **AI function calling**.
- Created automated content generation workflows with concern resolution and quality assurance processes.

Software Developer Intern **Feb 2024 - Jun 2024**
LiveSitter Pvt. Ltd. **Remote**

- Developed a Python & **Flask** server for decoding and processing frames from **RTSP** feeds from **500+ CCTV cameras**, providing **real-time insights**, and leveraged **Redis Pub/Sub** architecture to ensure scalability.
- Embedded post-inferencing performance monitoring with a human in the loop, increasing system reliability and reducing error rates by **30% in a month**.
- Built a **Redis** microservice for batch inferencing **yolo model** on camera frames, reducing GPU costs by **40%**, along with **CI/CD** and **AWS EC2 deployment**.

PROJECTS

Repel **Aug 2025 - Sep 2025**

- Built Repel, a full-stack web app demonstrating how LLMs can enable scalable **code generation** and execution for **broader projects**..
- Integrated an in-browser **code editor with Monaco** and executed code securely using **Web Containers**.

One-Shot Learning – Face Net Paper **Jan 2024 - May 2024**

- Built a custom architecture using **Inception layers** and **point-wise convolutions** to generate compact **face embeddings**.
- Implemented **Triplet Loss** to learn **similarity** metrics, **minimizing intra-class distance** and **maximizing inter-class separation**.

EDUCATION

B.Tech Computer Science Engineering (Spl - AIML) **Aug 2021 - Jul 2025**
University of Petroleum and Energy Studies, Dehradun

High School **Apr 2019 - Apr 2021**
DAV Public School, Faridabad